#This code will be used to demonstrate lemma 6.4. we demonstrate the existence of the curve γ1 in region A1.

#Construct the polynomial R from Theorem 6.1.3.

$$A := -(w+z)^3\left(\left(x^2y^2+z^4\right)w^4-x^2y^2z^4\right):$$

$$B := x^2y^5w^2z^2(w+z)^2(w-z)-x^2y^2(w+z)^3\left(w^2+z^2\right)w^2z^2(w-z)+w^4y^3z^4(w+z)^3 + w^4z^4x^3y^3:$$

$$R := B^2\cdot\left((w+z)^2-y^2\right)^3-A^2\cdot\left(y^2\cdot z\cdot w\right)^3$$

$$R := \left(x^2y^5w^2z^2(w+z)^2(w-z)-x^2y^2(w+z)^3\left(w^2+z^2\right)w^2z^2(w-z)+w^4y^3z^4(w+z)^3 + w^4z^4x^3y^3\right)^2\left((w+z)^2-y^2\right)^3-(w+z)^6\left(\left(x^2y^2+z^4\right)w^4-x^2y^2z^4\right)^2y^6z^3w^3 \quad \textbf{(1)}$$

#Apply the rational parametrization constructed in this chapter 4..

$$parametrization := \left\{ \mathrm{w}=\mathrm{a}+\frac{1}{a},\ z=b+\frac{1}{b},\ y=a-\frac{1}{a}+b-\frac{1}{b},\ x=a-\frac{1}{a}-b+\frac{1}{b}\right\}:$$

simplify(*numer*(expand(subs(parametrization, R)))) ;
denom(expand(subs(parametrization, R)));

$$-(a\,b-1)^6\Big(-4\,b^2\left(b^{28}+12\,b^{26}-155\,b^{24}+1018\,b^{22}+4315\,b^{20}+34120\,b^{18}+65514\,b^{16}+57358\,b^{14}+41514\,b^{12}+19656\,b^{10}+4187\,b^8+378\,b^6+37\,b^4+12\,b^2+1\right)a^{14}$$

$$-4\,b^2\left(b^{28}+12\,b^{26}+37\,b^{24}-518\,b^{22}-2085\,b^{20}+17992\,b^{18}+38058\,b^{16}+78414\,b^{14}+92394\,b^{12}+36360\,b^{10}+5211\,b^8-2118\,b^6-539\,b^4+76\,b^2+1\right)a^{18}+2\,b^4\left(b^{24}+36\,b^{22}+180\,b^{20}-2396\,b^{18}-25164\,b^{16}-42552\,b^{14}-90894\,b^{12}-84792\,b^{10}-12876\,b^8-348\,b^6+3252\,b^4+420\,b^2-127\right)a^{20}+8\Big(b^{20}+b^{18}-27\,b^{16}-159\,b^{14}+\frac{2163}{2}\,b^{12}-1827\,b^{10}-\frac{1037}{2}\,b^8+2369\,b^6-699\,b^4+b^2+1\Big)b^6a^{22}-1024\,b^{15}\left(b^2+1\right)^4a^7$$

$$+ 2 b^4 \left(b^{24} + 36 b^{22} - 2636 b^{20} - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12}\right.$$

$$\left. - 25400 b^{10} - 9036 b^8 - 1372 b^6 + 180 b^4 + 36 b^2 + 1\right) a^{12} - 9 \left(b^{16} + 8 b^{14} - 16 b^{12}\right.$$

$$\left. - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 - 528 b^4 + \frac{4168}{9} b^2 + 1\right) b^8 a^{24} + \left(-4 b^{18} + 8 b^{16}\right.$$

$$\left. - 4 b^{14}\right) a^{30} + \left(6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}\right) a^{28} + \left(72 b^{20} - 76 b^{18} + 8 b^{16}\right.$$

$$\left. + 692 b^{14} + 72 b^{12} - 1024 b^{10}\right) a^{26} + \left(b^{32} + 8 b^{30} + 20 b^{28} - 120 b^{26} - 3426 b^{24} - 7288 b^{22}\right.$$

$$\left. - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} - 157780 b^{12} - 32376 b^{10} + 1950 b^8\right.$$

$$\left. + 1672 b^6 + 20 b^4 + 8 b^2 + 1\right) a^{16} + \left(-1016 b^{26} - 8440 b^{24} - 21976 b^{22} - 25848 b^{20}\right.$$

$$\left. - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} + 8 b^8 + 8 b^6\right) a^{10} + \left(-265 b^{24}\right.$$

$$\left. - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} + 144 b^{12} - 72 b^{10} - 9 b^8\right) a^8 + \left(\right.$$

$$\left. -512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} + 72 b^{12}\right) a^6 + \left(-250 b^{20} - 288 b^{18}\right.$$

$$\left. + 44 b^{16} - 32 b^{14} + 6 b^{12}\right) a^4 + \left(-4 b^{18} + 8 b^{16} - 4 b^{14}\right) a^2 + a^{32} b^{16} - 1536 \left(b^8 - b^6\right.$$

$$\left. + \frac{5}{3} b^4 - \frac{16}{3} b^2 + \frac{5}{3}\right) \left(b^2 + 1\right)^4 b^9 a^{13} - 2560 \left(b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 - 1\right) \left(b^2\right.$$

$$\left. + 1\right)^4 b^{11} a^{11} - 1024 \left(b^2 - \frac{1}{2}\right) \left(b^2 + 1\right)^5 b^{13} a^9 + 512 b^7 \left(b^{10} - 22 b^8 + 81 b^6 - 78 b^4\right.$$

$$\left. + 11 b^2 + 1\right) \left(b^2 + 1\right)^4 a^{17} + 512 b^7 \left(b^{10} - 19 b^8 + 42 b^6 - 9 b^4 - 10 b^2 + 1\right) \left(b^2 + 1\right)^4 a^{15}$$

$$+ 5632 \left(b^2 + 1\right)^4 \left(b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11}\right) b^7 a^{21} - 3584 \left(b^2 + 1\right)^4 \left(b^8 - \frac{52}{7} b^6\right.$$

$$\left. + \frac{95}{7} b^4 - \frac{43}{7} b^2 + \frac{3}{7}\right) b^7 a^{19} - 2560 \left(b^2 + 1\right)^4 \left(b^2 - 2\right) \left(b^2 - \frac{1}{5}\right) b^7 a^{23} + b^{16}\Big)$$

$$\left(a b + 1\right)^6 \left(b^2 + 1\right)^3 \left(a + b\right)^{12} \left(a^2 + 1\right)^3$$

$$a^{31} b^{31} \tag{2}$$

#Define r1

$$r1 := \Big(b^{16} - 4 b^2 \left(b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16}\right.$$

$$+ 57358\, b^{14} + 41514\, b^{12} + 19656\, b^{10} + 4187\, b^{8} + 378\, b^{6} + 37\, b^{4} + 12\, b^{2} + 1\big)\, a^{14} + 2\, b^{4}\, \big(b^{24}$$

$$+ 36\, b^{22} - 2636\, b^{20} - 6620\, b^{18} - 27212\, b^{16} - 64184\, b^{14} - 54798\, b^{12} - 25400\, b^{10} - 9036\, b^{8}$$

$$- 1372\, b^{6} + 180\, b^{4} + 36\, b^{2} + 1\big)\, a^{12} - 1024\, b^{15}\, \big(b^{2} + 1\big)^{4}\, a^{7} - 4\, b^{2}\, \big(b^{28} + 12\, b^{26} + 37\, b^{24}$$

$$- 518\, b^{22} - 2085\, b^{20} + 17992\, b^{18} + 38058\, b^{16} + 78414\, b^{14} + 92394\, b^{12} + 36360\, b^{10} + 5211\, b^{8}$$

$$- 2118\, b^{6} - 539\, b^{4} + 76\, b^{2} + 1\big)\, a^{18} + 2\, b^{4}\, \big(b^{24} + 36\, b^{22} + 180\, b^{20} - 2396\, b^{18} - 25164\, b^{16}$$

$$- 42552\, b^{14} - 90894\, b^{12} - 84792\, b^{10} - 12876\, b^{8} - 348\, b^{6} + 3252\, b^{4} + 420\, b^{2} - 127\big)\, a^{20}$$

$$+ 8\, b^{6}\, \Big(b^{20} + b^{18} - 27\, b^{16} - 159\, b^{14} + \frac{2163}{2}\, b^{12} - 1827\, b^{10} - \frac{1037}{2}\, b^{8} + 2369\, b^{6} - 699\, b^{4}$$

$$+ b^{2} + 1\Big)\, a^{22} - 9\, b^{8}\, \Big(b^{16} + 8\, b^{14} - 16\, b^{12} - 72\, b^{10} + \frac{3698}{9}\, b^{8} - 328\, b^{6} - 528\, b^{4} + \frac{4168}{9}\, b^{2}$$

$$+ 1\Big)\, a^{24} + \big(-4\, b^{18} + 8\, b^{16} - 4\, b^{14}\big)\, a^{30} + \big(6\, b^{20} - 32\, b^{18} + 44\, b^{16} - 32\, b^{14} + 6\, b^{12}\big)\, a^{28}$$

$$+ \big(72\, b^{20} - 76\, b^{18} + 8\, b^{16} + 692\, b^{14} + 72\, b^{12} - 1024\, b^{10}\big)\, a^{26} + \big(b^{32} + 8\, b^{30} + 20\, b^{28}$$

$$- 120\, b^{26} - 3426\, b^{24} - 7288\, b^{22} - 125524\, b^{20} - 222104\, b^{18} - 279672\, b^{16} - 317080\, b^{14}$$

$$- 157780\, b^{12} - 32376\, b^{10} + 1950\, b^{8} + 1672\, b^{6} + 20\, b^{4} + 8\, b^{2} + 1\big)\, a^{16} + \big(-1016\, b^{26}$$

$$- 8440\, b^{24} - 21976\, b^{22} - 25848\, b^{20} - 37428\, b^{18} - 31000\, b^{16} - 8756\, b^{14} + 264\, b^{12} - 216\, b^{10}$$

$$+ 8\, b^{8} + 8\, b^{6}\big)\, a^{10} + \big(-265\, b^{24} - 1864\, b^{22} + 2192\, b^{20} + 4232\, b^{18} + 142\, b^{16} - 1912\, b^{14}$$

$$+ 144\, b^{12} - 72\, b^{10} - 9\, b^{8}\big)\, a^{8} + \big(-512\, b^{22} - 696\, b^{20} + 692\, b^{18} + 1288\, b^{16} - 76\, b^{14}$$

$$+ 72\, b^{12}\big)\, a^{6} + \big(-250\, b^{20} - 288\, b^{18} + 44\, b^{16} - 32\, b^{14} + 6\, b^{12}\big)\, a^{4} + \big(-4\, b^{18} + 8\, b^{16}$$

$$- 4\, b^{14}\big)\, a^{2} + a^{32}\, b^{16} - 2560\, \Big(b^{2} - \frac{1}{5}\Big)\, b^{7}\, \big(b^{2} + 1\big)^{4}\, \big(b^{2} - 2\big)\, a^{23} - 1536\, \Big(b^{8} - b^{6} + \frac{5}{3}\, b^{4}$$

$$- \frac{16}{3}\, b^{2} + \frac{5}{3}\Big)\, b^{9}\, \big(b^{2} + 1\big)^{4}\, a^{13} - 2560\, \Big(b^{6} - \frac{8}{5}\, b^{4} + \frac{6}{5}\, b^{2} - 1\Big)\, b^{11}\, \big(b^{2} + 1\big)^{4}\, a^{11}$$

$$- 1024\, b^{13}\, \big(b^{2} + 1\big)^{5}\, \Big(b^{2} - \frac{1}{2}\Big)\, a^{9} + 512\, b^{7}\, \big(b^{10} - 22\, b^{8} + 81\, b^{6} - 78\, b^{4} + 11\, b^{2}$$

$$+ 1\big)\, \big(b^{2} + 1\big)^{4}\, a^{17} + 512\, b^{7}\, \big(b^{10} - 19\, b^{8} + 42\, b^{6} - 9\, b^{4} - 10\, b^{2} + 1\big)\, \big(b^{2} + 1\big)^{4}\, a^{15}$$

$$+ 5632\, b^{7}\, \big(b^{2} + 1\big)^{4}\, \Big(b^{6} - \frac{42}{11}\, b^{4} + \frac{38}{11}\, b^{2} - \frac{5}{11}\Big)\, a^{21} - 3584\, \Big(b^{8} - \frac{52}{7}\, b^{6} + \frac{95}{7}\, b^{4}$$

$$- \frac{43}{7}\, b^{2} + \frac{3}{7}\Big)\, b^{7}\, \big(b^{2} + 1\big)^{4}\, a^{19}\Big) :$$

$collect(expand(r1), a)$

$$(-4b^{18}+8b^{16}-4b^{14})a^{30}+(6b^{20}-32b^{18}+44b^{16}-32b^{14}+6b^{12})a^{28}+(72b^{20}-76b^{18}+8b^{16}+692b^{14}+72b^{12}-1024b^{10})a^{26}+(b^{32}+8b^{30}+20b^{28}-120b^{26}-3426b^{24}-7288b^{22}-125524b^{20}-222104b^{18}-279672b^{16}-317080b^{14}-157780b^{12}-32376b^{10}+1950b^{8}+1672b^{6}+20b^{4}+8b^{2}+1)a^{16}+(-1016b^{26}-8440b^{24}-21976b^{22}-25848b^{20}-37428b^{18}-31000b^{16}-8756b^{14}+264b^{12}-216b^{10}+8b^{8}+8b^{6})a^{10}+(-265b^{24}-1864b^{22}+2192b^{20}+4232b^{18}+142b^{16}-1912b^{14}+144b^{12}-72b^{10}-9b^{8})a^{8}+(-512b^{22}-696b^{20}+692b^{18}+1288b^{16}-76b^{14}+72b^{12})a^{6}+(-250b^{20}-288b^{18}+44b^{16}-32b^{14}+6b^{12})a^{4}+(-4b^{18}+8b^{16}-4b^{14})a^{2}+a^{32}b^{16}+(-9b^{24}-72b^{22}+144b^{20}+648b^{18}-3698b^{16}+2952b^{14}+4752b^{12}-4168b^{10}-9b^{8})a^{24}+(-2560b^{19}-4608b^{17}+6144b^{15}+19456b^{13}+13824b^{11}+1536b^{9}-1024b^{7})a^{23}+(8b^{26}+8b^{24}-216b^{22}-1272b^{20}+8652b^{18}-14616b^{16}-4148b^{14}+18952b^{12}-5592b^{10}+8b^{8}+8b^{6})a^{22}+(5632b^{21}+1024b^{19}-32768b^{17}-31232b^{15}+26112b^{13}+40960b^{11}+9216b^{9}-2560b^{7})a^{21}+(2b^{28}+72b^{26}+360b^{24}-4792b^{22}-50328b^{20}-85104b^{18}-181788b^{16}-169584b^{14}-25752b^{12}-696b^{10}+6504b^{8}+840b^{6}-254b^{4})a^{20}+(-3584b^{23}+12288b^{21}+36352b^{19}-27136b^{17}-102400b^{15}-41984b^{13}+30208b^{11}+15872b^{9}-1536b^{7})a^{19}+(-4b^{30}-48b^{28}-148b^{26}+2072b^{24}+8340b^{22}-71968b^{20}-152232b^{18}-313656b^{16}-369576b^{14}-145440b^{12}-20844b^{10}+8472b^{8}+2156b^{6}-304b^{4}-4b^{2})a^{18}+(512b^{25}-9216b^{23}-512b^{21}+60416b^{19}+50176b^{17}-61952b^{15}-82432b^{13}-14336b^{11}+7680b^{9}+512b^{7})a^{17}+(512b^{25}-7680b^{23}-14336b^{21}+25088b^{19}+67072b^{17}+28672b^{15}-25600b^{13}-22016b^{11}-3072b^{9}+512b^{7})a^{15}+(-4b^{30}-48b^{28}+620b^{26}-4072b^{24}-17260b^{22}-136480b^{20}-262056b^{18}-229432b^{16}-166056b^{14}-78624b^{12}-16748b^{10}-1512b^{8}-148b^{6}-48b^{4}-4b^{2})a^{14}+(-1536b^{25}-4608b^{23}-5632b^{21}+1024b^{19}+19456b^{17}+30208b^{15}+14848b^{13}-2048b^{11}-2560b^{9})a^{13}+(2b^{28}+72b^{26}-5272b^{24}-13240b^{22}-54424b^{20}-128368b^{18}-109596b^{16}-50800b^{14}-18072b^{12}-2744b^{10}+360b^{8}+72b^{6}+2b^{4})a^{12}+(-2560b^{25}-6144b^{23}-2048b^{21}+4608b^{19}+5632b^{17}+7168b^{15}+7168b^{13}+2560b^{11})a^{11}+(-1024b^{25}-4608b^{23}-7680b^{21}-5120b^{19}+1536b^{15}+512b^{13})a^{9}+(-1024b^{23}-4096b^{21}-6144b^{19}-4096b^{17}-1024b^{15})a^{7}+b^{16} \quad \textbf{(3)}$$

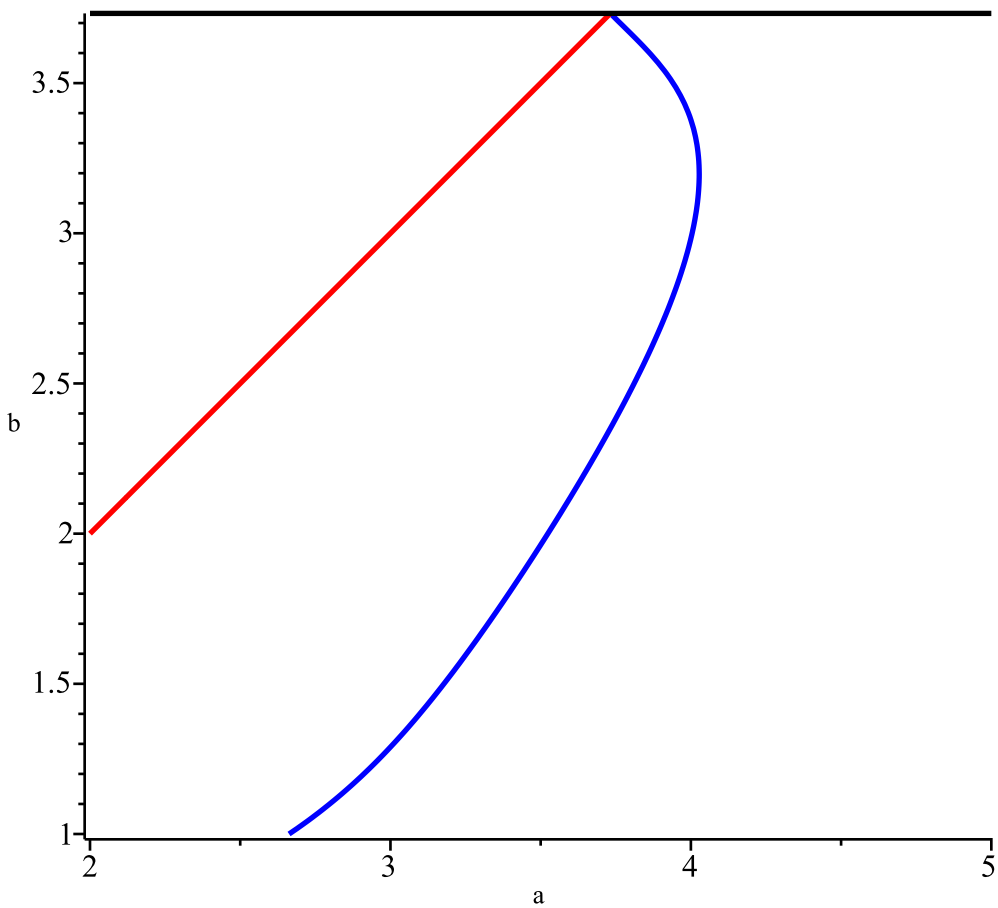
#Plot the curve r1=0 in region A1.

$r := a - b :$
$rr := b - 2 - \mathrm{sqrt}(3) :$

$\mathrm{with}(\mathrm{plots}) :$

```
implicitplot( [r1 = 0, r = 0, rr = 0], a = 2 .. 5, b = 1 .. (2 + sqrt(3)),
        scaling = constrained,
        gridrefine = 2,
        thickness = 2,
        color = [blue, red, black],
        labels = ["a", "b"],
         title = "Graphic for r1=0 in A1");
```

Graphic for r1=0 in A1



#Apply the Möbius transformation on the rectangle A1 = [2, 5) × $[1, 2 + \mathrm{sqrt}(3))$ *constructing r̂1*

$$moebius := \left\{ a = \frac{2 + 5\,c}{1 + c},\ b = \frac{1 + (2 + \mathrm{sqrt}(3))\,d}{1 + d} \right\} :$$

$$\hat{r}1 := \mathrm{collect}(\mathrm{numer}(\mathrm{expand}(\mathrm{subs}(moebius, r1))), c) :$$

#Compute the degree of r̂1 in c

$degree(\hat{r}1, c);$

$$32 \tag{4}$$

#Define the coefficients ci = coeff (r̂1, c, i)

$seq(evalf(assign(cat('c', i), coeff(\hat{r}1, c, i))), i = 0 .. 32);$

#Compute the all degree of ci in d
$seq(degree(eval(cat('c', i)), d), i = 0 .. 32);$

$$32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32 \tag{5}$$

#From here we will analyze the signal variation of all coefficients ci

$seq(sign(evalf(coeff(c32, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{6}$$

$seq(sign(evalf(coeff(c31, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{7}$$

$seq(sign(evalf(coeff(c30, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{8}$$

$seq(sign(evalf(coeff(c29, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{9}$$

$seq(sign(evalf(coeff(c28, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{10}$$

$seq(sign(evalf(coeff(c27, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{11}$$

$seq(sign(evalf(coeff(c26, d, i))), i = 32 .. 0, -1);$

#Prove that c26 has no positive root

$sign(evalf(coeff(c26, d, 26) + coeff(c26, d, 29)));$
$sign(evalf(coeff(c26, d, 25) + coeff(c26, d, 28)));$
$sign(evalf(coeff(c26, d, 24) + coeff(c26, d, 27)));$

$sign\left(evalf\left(coeff(c26, d, 29)^2 - 4\, coeff(c26, d, 32) coeff(c26, d, 26)\right)\right);$
$sign\left(evalf\left(coeff(c26, d, 28)^2 - 4\, coeff(c26, d, 31) coeff(c26, d, 25)\right)\right);$

$sign\left(evalf\left(coeff(c26, d, 27)^2 - 4\, coeff(c26, d, 30)\, coeff(c26, d, 24)\right)\right);$

$$1, 1, 1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$1$$

$$1$$

$$1$$

$$-1$$

$$-1$$

$$-1 \tag{12}$$

$seq(sign(evalf(coeff(c25, d, i))), i = 32 .. 0, -1);$
$fsolve(c25 = 0, d, 0 .. \text{infinity});$

$$1, 1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$1.564508722, 10.90397742 \tag{13}$$

$seq(sign(evalf(coeff(c24, d, i))), i = 32 .. 0, -1);$
$fsolve(c24 = 0, d, 0 .. \text{infinity});$

$$1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$1.055336702, 23.20521862 \tag{14}$$

$seq(sign(evalf(coeff(c23, d, i))), i = 32 .. 0, -1);$
$fsolve(c23 = 0, d, 0 .. \text{infinity});$

$$1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$0.7717026565, 103.9394695 \tag{15}$$

$seq(sign(evalf(coeff(c22, d, i))), i = 32 .. 0, -1);$
$fsolve(c22 = 0, d, 0 .. \text{infinity});$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$0.5827251306 \tag{16}$$

$seq(sign(evalf(coeff(c21, d, i))), i = 32 .. 0, -1);$
$fsolve(c21 = 0, d, 0 .. \text{infinity});$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$0.4458863161 \tag{17}$$

$seq(sign(evalf(coeff(c20, d, i))), i = 32 .. 0, -1);$
$fsolve(c20 = 0, d, 0 .. \text{infinity});$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$$

$$0.3421854334 \tag{18}$$

$seq(sign(evalf(coeff(c19, d, i))), i = 32 .. 0, -1);$

$fsolve(c19 = 0, d, 0 .. \text{infinity});$

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$

$$0.2615891332 \tag{19}$$

$seq(sign(evalf(coeff(c18, d, i))), i = 32 .. 0, -1);$
$fsolve(c18 = 0, d, 0 .. \text{infinity});$

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1$

$$0.1981073283 \tag{20}$$

$seq(sign(evalf(coeff(c17, d, i))), i = 32 .. 0, -1);$
$fsolve(c17 = 0, d, 0 .. \text{infinity});$

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1$

$$0.1477275781 \tag{21}$$

$seq(sign(evalf(coeff(c16, d, i))), i = 32 .. 0, -1);$
$fsolve(c16 = 0, d, 0 .. \text{infinity});$

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1$

$$0.1074816816 \tag{22}$$

$seq(sign(evalf(coeff(c15, d, i))), i = 32 .. 0, -1);$
$fsolve(c15 = 0, d, 0 .. \text{infinity});$

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1$

$$0.07504800351 \tag{23}$$

$seq(sign(evalf(coeff(c14, d, i))), i = 32 .. 0, -1);$
$fsolve(c14 = 0, d, 0 .. \text{infinity});$

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1$

$$0.04859391073 \tag{24}$$

$seq(sign(evalf(coeff(c13, d, i))), i = 32 .. 0, -1);$
$fsolve(c13 = 0, d, 0 .. \text{infinity});$

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1$

$$0.02669523547 \tag{25}$$

$seq(sign(evalf(coeff(c12, d, i))), i = 32 .. 0, -1);$
$fsolve(c12 = 0, d, 0 .. \text{infinity});$

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1$

$$0.008266650534 \tag{26}$$

$seq(sign(evalf(coeff(c11, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(27)}$$

$seq(sign(evalf(coeff(c10, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(28)}$$

$seq(sign(evalf(coeff(c9, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(29)}$$

$seq(sign(evalf(coeff(c8, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(30)}$$

$seq(sign(evalf(coeff(c7, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(31)}$$

$seq(sign(evalf(coeff(c6, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(32)}$$

$seq(sign(evalf(coeff(c5, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(33)}$$

$seq(sign(evalf(coeff(c4, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(34)}$$

$seq(sign(evalf(coeff(c3, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(35)}$$

$seq(sign(evalf(coeff(c2, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(36)}$$

$seq(sign(evalf(coeff(c1, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(37)}$$

$seq(sign(evalf(coeff(c0, d, i))), i = 32 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(38)}$$

#Let's verify that the γ1 curve found in A1 is completely contained within the region of the R2 parameters.

$simplify(eval(r1, a = b))$

$$b^{16}\left(b^2-4b+1\right)\left(b^2+4b+1\right)\left(b^4+4b^3+18b^2+4b+1\right)\left(b^4-4b^3+18b^2-4b+1\right)\left(b^2+1\right)^{10} \tag{39}$$

$sturm\left(sturmseq\left(b^4-4b^3+18b^2-4b+1, b\right), b, 1, 2+\mathrm{sqrt}(3)\right);$

$$0 \tag{40}$$

$solve\left(b^2-4b+1=0\right)$

$$2+\sqrt{3}, 2-\sqrt{3} \tag{41}$$